Expanding Brackets

Exam code: 9MA0

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Expanding brackets

How do I expand brackets?

- To expand brackets, the rule is that each term in one set of brackets must be multiplied by each term in the other set of brackets

$$(a+b)(x+y+z) = a(x+y+z) + b(x+y+z)$$
$$= ax + ay + az + bx + by + bz$$

$$(a+b+c)(x+y+z) = a(x+y+z) + b(x+y+z) + c(x+y+z)$$
$$= ax + ay + az + bx + by + bz + cx + cy + cz$$

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- 'FOIL' is a special case of this, when each set of brackets contains two terms you can multiply in order
  - e.g. to multiply out or expand these brackets (2x + 3)(x - 5)
    - First (**2x** + 3)(**x** - 5) gives $2x^2$
    - Outside (**2x** + 3)(x **- 5**) gives -10x
    - Inside (2x + **3**)(**x** - 5) gives +3x

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

brackets out twice or three times respectively
  - e.g. $(3x + 2)^2 = (3x + 2)(3x + 2) = 9x^2 + 6x + 6x + 4 = 9x^2 + 12x + 4$

- If you are trying to expand something like $(a + b)^n$ for powers of n greater than 2 or 3, use the **binomial expansion**
- If you have to expand more than two sets of brackets, just expand them two at a time

Worked Example

Expand and simplify $4x(x - 3y)(3x + 2y - 5)$.

EXPAND $4x(x-3y)$ FIRST

$4x(x-3y) = 4x^2 - 12xy$

MULTIPLY EACH TERM IN THE FIRST BRACKET BY EACH TERM IN THE SECOND BRACKET

$(4x^2 - 12xy)(3x + 2y - 5)$

$= 12x^3 + 8x^2y - 20x^2 - 36x^2y - 24xy^2 + 60xy$

$= 12x^3 - 28x^2y - 20x^2 - 24xy^2 + 60xy$

COLLECT LIKE TERMS

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Worked Example

Expand $(x - 4)(x + 3)(x^2 + x - 1)$

FIRST EXPAND $(x-4)(x+3) = x^2 - x - 12$

THEN CONTINUE: $(x^2 - x - 12)(x^2 + x - 1)$

$= x^2(x^2 + x - 1) - x(x^2 + x - 1) - 12(x^2 + x - 1)$

$= x^4 + x^3 - x^2 - x^3 - x^2 + x - 12x^2 - 12x + 12$

$= x^4 - 14x^2 - 11x + 12$

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